CS236-Project Phase 1

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Source Code:

[1] Schema mapping: DBMS

[2] Source code: https://github.com/LydiaNiu/CS236-database-management

[3] Video presentation: [link]

01. Dataset Objective

This project uses 2 datasets, the New York City 311 and Los Angeles MyLA311 datasets to explore how residents report service requests and how those requests are handled in the corresponding cities. Each dataset contains 800,000 records, giving us a combined total of 1,600,000 requests. The NYC dataset covers July 22 to October 5, 2026 while the LA dataset covers January 1 to July 24, 2024, based on the request creation dates. NYC contains 44 source columns, while LA only has 34. Each dataset also has its own unique identifier: unique_key for NYC and srnumber for LA.

Both datasets contain information such as request types, submission channels, statuses, locations(addresses), creation and closure dates, and longitude and latitude. However, they use different schemas and categories, so our objective is to clean and standardize these differences, then integrate the two datasets into a consistent, unified dataset while preserving useful information from each source.

At the end of this phase, we will finish the exploratory data analysis (EDA), data cleaning, and data integration.

02. Exploratory Data Analysis

Duplicate Records

Check	NYC	LA	Result
Total rows	800000	800000	
Null IDs	0	0	Pass
Distinct IDs	800000	800000	Pass
Duplicate IDs	0	0	Pass
IDs with invalid format	0	0	Pass
Duplicate rows (all columns)	0	0	Pass
Duplicate groups (ignoring ID)	217	0	Review
Surplus duplicate rows (ignoring ID)	261	0	Review

Both datasets have the same number of records (800,000), all with unique ID's, serving as a primary key for their respective datasets. To check whether duplicates existed with a different ID (as maybe the requester submitted the exact same request twice, or the ticketing system created two duplicate records with different IDs). LA maintains 0 duplicates, while NYC

has 261 surplus records (0.03%) that match another record in every other column, forming 217 duplicate groups. The duplicates are mostly concentrated in a few complaint types (HPD/General, DOT/Street Condition, and DEP/Air quality mostly). We can assume, but cannot confirm due to the data, that a single submission is generating multiple tickets for certain types. If the errors were indeed random, we might see a more equal distribution of duplicate tickets across all types. Because we cannot confirm the cause of these duplicate (which do, in fact, have different unique keys) records, we can leave them in the dataset, as there is also not a substantial amount of them (only 0.03%), and they don't threaten the integrity of the data.

Agency / Complaint Type	Rows	Share
HPD / GENERAL	185	38.70%
DOT / Street Condition	69	14.40%
DEP / Air Quality	54	11.30%
HPD / UNSANITARY CONDITION	44	9.20%
DOT / Street Light Condition	40	8.40%
HPD / HEAT/HOT WATER	12	2.50%
HPD / PLUMBING	10	2.10%
DPR / Root/Sewer/Sidewalk Condition	9	1.90%
DOT / Traffic Signal Condition	8	1.70%
DOB / Building/Use	8	1.70%
All other types	39	8.20%

Date Analysis:

The two datasets cover different time periods. The NYC 311 dataset goes from July 22, 2026, to October 5, 2026, while the Los Angeles Dataset goes from January 1, 2024, to July 24, 2024 (considering only the created date). Their lack of overlap will somewhat limit the analysis that we can do, as we cannot consider their relationship across the same period of time. However, they overlap partially in the months they are covering, even if in different years; meaning we can still analyze their relationship in that manner.

Dataset	min_created	max_created	min_closed	max_closed
NYC	2026-07-22 9:27:15	2026-10-05 2:05:08	2026-07-08 23:36:00	2026-10-05 3:27:00
LA	2024-01-01 0:00:06	2024-07-24 23:14:35	2024-01-01 7:39:53	2025-10-16 0:02:44

Anomalies in the Date and Zip Code.

Date and Zip Code Anomalies	NYC	LA
Closed before created	626	0
Negative resolution times	626	0
Invalid ZIP (not 5 digits)	1	39

Here, we can see that there are 626 records in the NYC dataset that were closed before they were created (leading to 626 negative resolution times). This could be due to minor systematic errors across agencies and the ticketing system for 311. The 626 records are only about 0.08% of all NYC records. There appear to be some anomalies in zip codes as well, with 39 records in the LA dataset having invalid zip codes, and 1 in NYC.

Dataset	zip_value	count
NYC	No	1
LA	0	38
LA	VE, 0	1

NYC has a placeholder value of No for zip code, while LA primarily uses 0 as a placeholder for Null, with a single instance of `VE, 0`.

Testing for Null Placeholders

Placeholders: ('', 'unspecified', 'n/a', 'unknown', 'none')

Testing amongst common null placeholders leads to some results in `status`(53 records), and a comparably high amount in `open_data_channel_type (65687)

Dataset	Column	Placeholder_Count
NYC	status	53
NYC	open_data_channel_type	65687
NYC	complaint_type	0
NYC	agency	0
LA	status	0
LA	requestsource	0
LA	requesttype	0

Location Checks

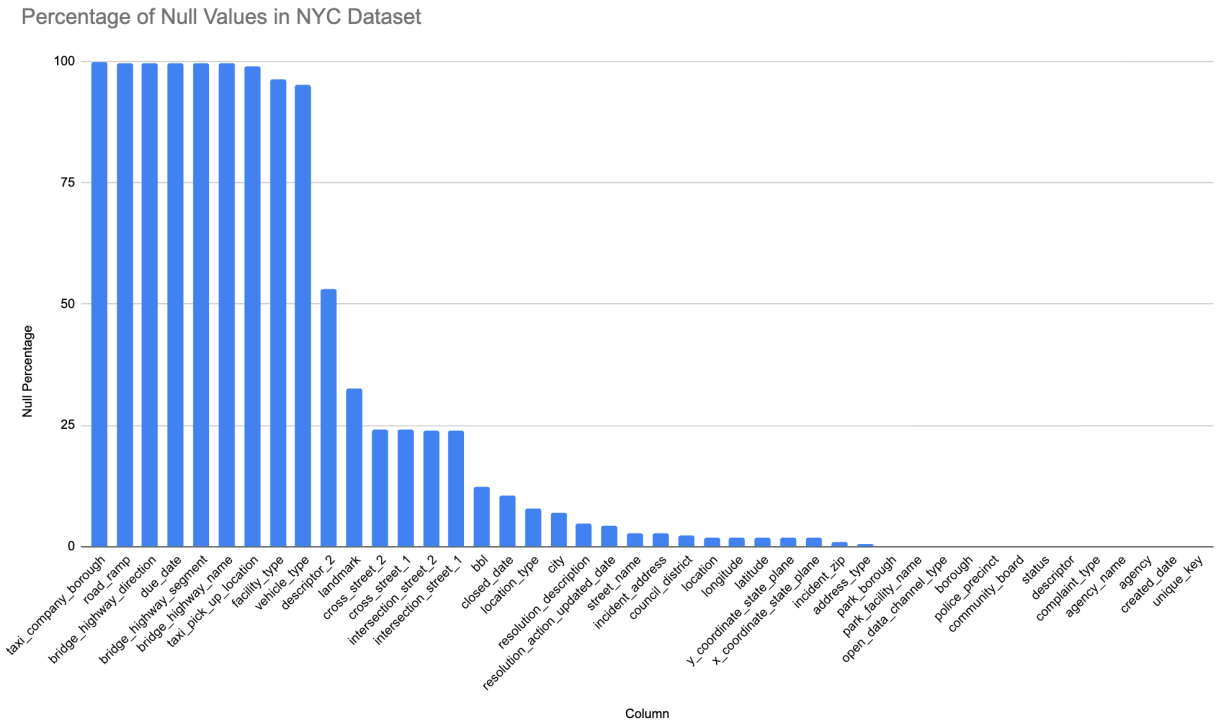
Summary statistics for location demonstrate no outliers outside of their respective cities. Min-max ranges stay within the boundaries of their respective locations.

Statistic	NYC_latitude	NYC_longitude	LA_latitude	LA_longitude
count	785370	785370	799233	799233
mean	40.7311	-73.9222	34.0896	-118.3624
stddev	0.0846	0.0778	0.1197	0.1102
min	40.4989	-74.2549	33.7061	-118.6678
max	40.9129	-73.7004	34.3343	-118.1558

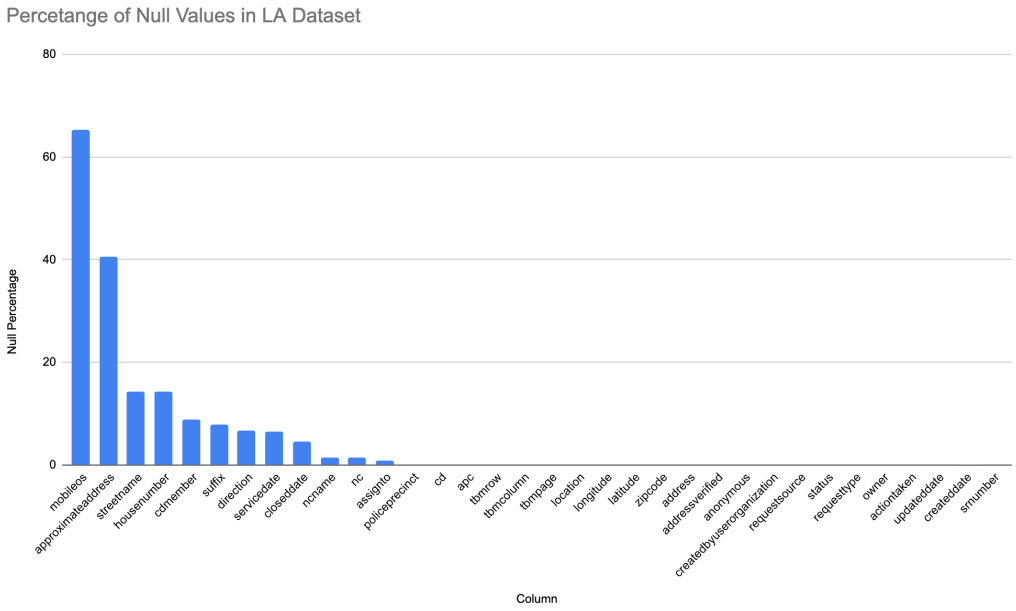
Null Value Percentages

Null percentages are exceedingly high (>90%) in some columns for the NYC dataset. These columns are`facility_type`, `due_date`, `vehicle_type`, `taxi_company_borough`, `taxi_pick_up_location`, `bridge_highway_segment`, `bridge_highway_name`, `bridge_highway_direction`, `road_ramp`. These columns pertain to only a specific and

uncommon instance of 311 incidents, leading to a high null percentage. Facility_type only carries two values: Null and DSNY Garage. Due_date only has ~3,000 instances apart from null, carrying little information. The other variables carry more structural information than the former two variables, and they apply only to specific complaint types (taxi, bridge/highway), so the nulls are structural rather than data-quality problems.

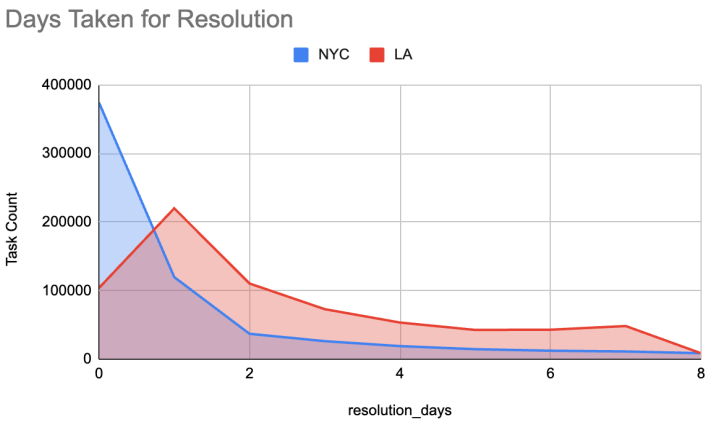


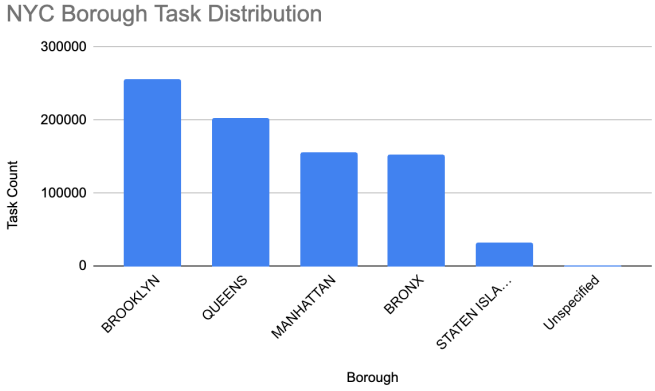
Null values in the LA dataset are seemingly less problematic. The top missing value, `mobileos`, applies only to incidents reported through mobile devices and records their operating system (either iOS or Android), otherwise marking it as Null. This is proven by the fact that ~34% of the variable is filled, matching the percentage of requestsource = Mobile App in the dataset. The variables `streename`, and `housenumber` are rendered redundant by address, while `approximateadress` is a binary variable denoting if the address was approximated. The rest of the variables with null values are below the 10% threshold, posing little threat to data integrity.



Days Taken for Resolution

The pattern for resolution is similar across the two datasets. A majority of the incidents are resolved in the first few days, following a general downward trend as time passes. LA differs from NYC, having its peak of resolved incidents on its second day, rather than the first.

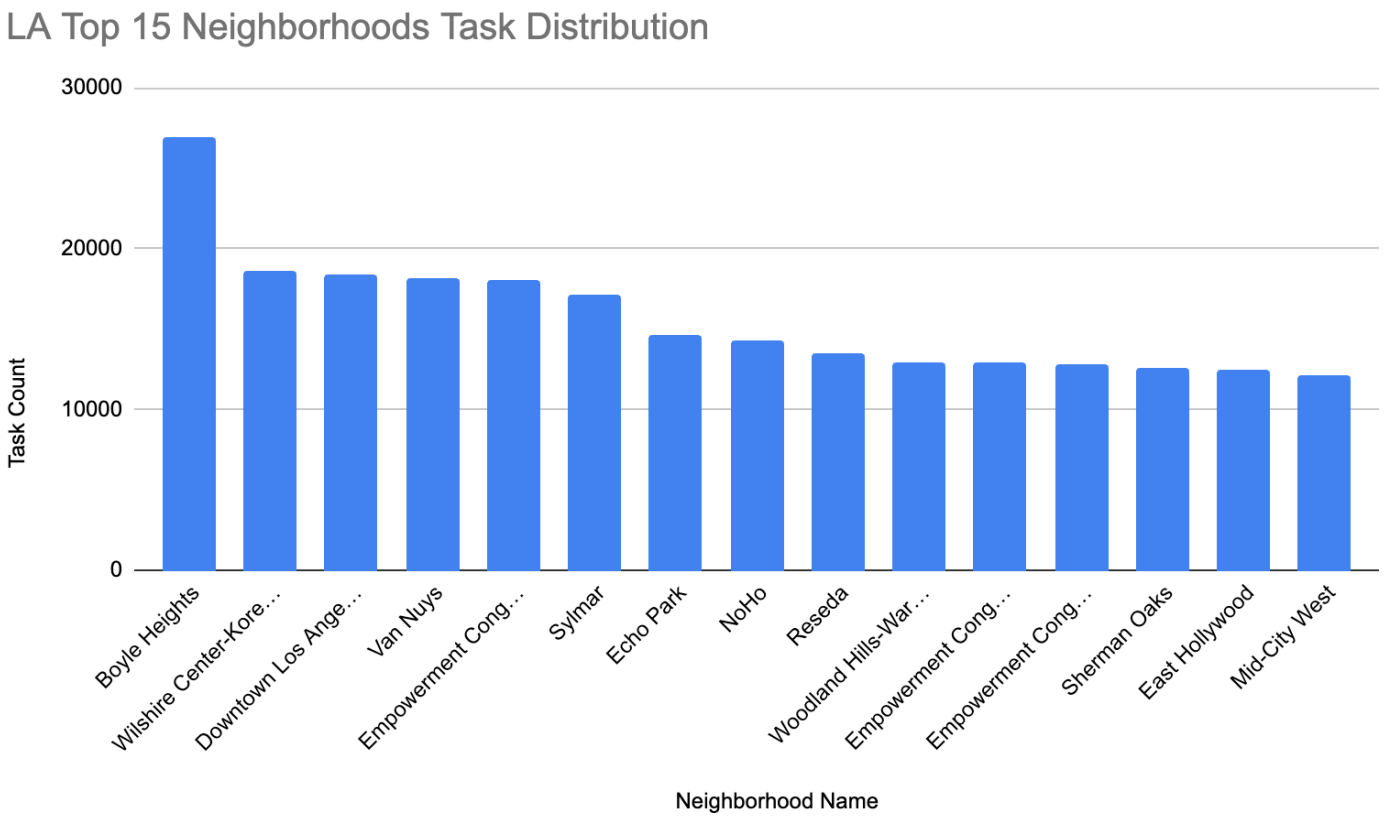




Neighborhood and Borough Distributions

New York City categorizes its records based on boroughs, rather than neighborhoods. A total of only 5 boroughs leaves a relatively simple distribution, with Brooklyn having the largest number of recorded tuples. Note that missing values for boroughs are incredibly low.

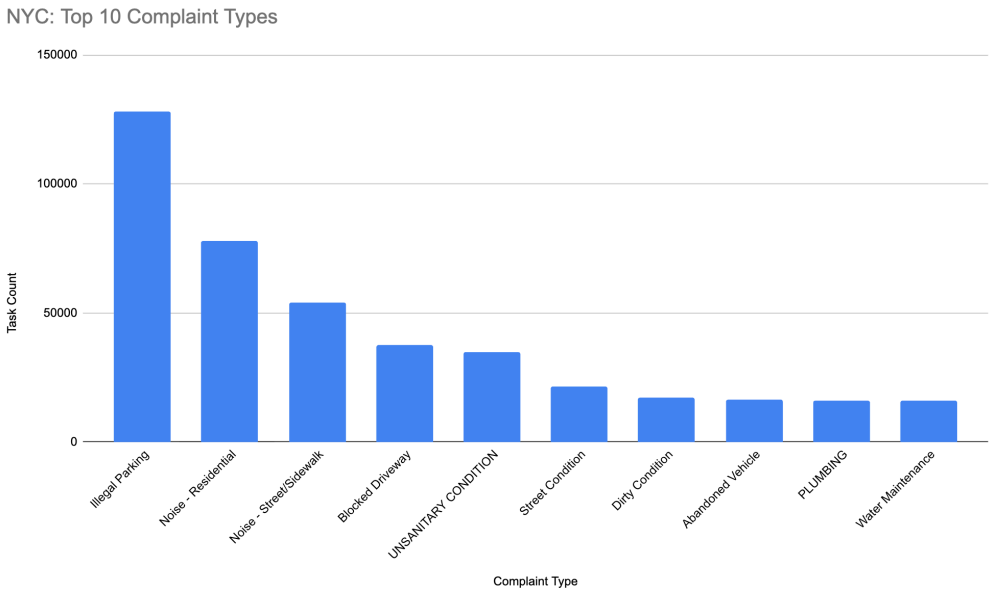
Los Angeles, on the other hand, categorizes based on Neighborhood. Due to the high number of neighborhoods, we can extract the top 15 neighborhoods. Boyle Heights (not the most populated, nor the biggest neighborhood) seems to have a relatively high number of records allotted to it, while the rest of the neighborhoods steadily decline in task count.

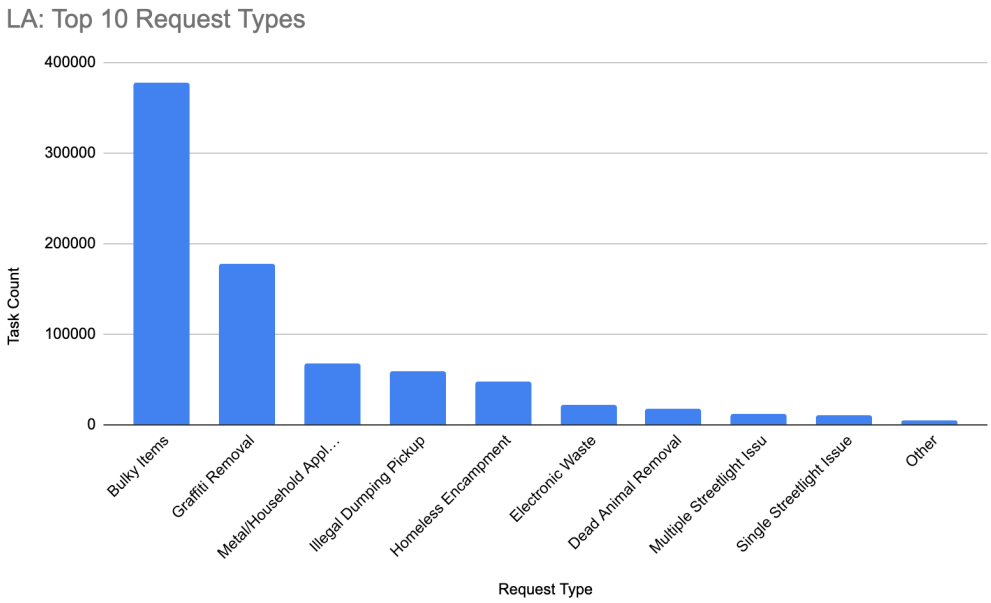


Complaint/Request Types

The top complaint and request types demonstrate that there is very little overlap in the type of complaints reported between NYC and LA. However, the possible categories for NYC are 184. There are many that overlap with LA; they just simply rank too low in task counts. Some possible matches include ("Graffiti Removal", "Graffiti"), ("Homeless

Encampment", "Encampment"), ("Dead Animal Removal", "Dead Animal"), ("Single/Multiple Streetlight Issue", "Street Light Condition"), ("Illegal Dumping Pickup", "Illegal Dumping").





Status

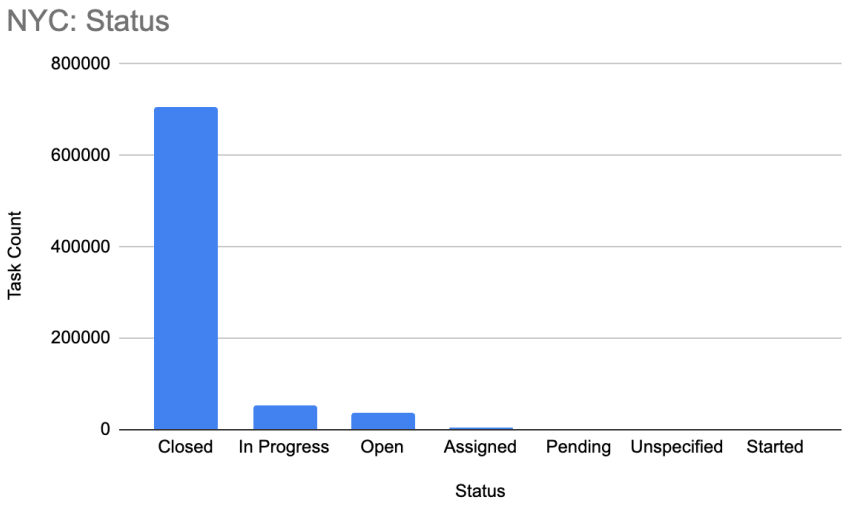
Distribution

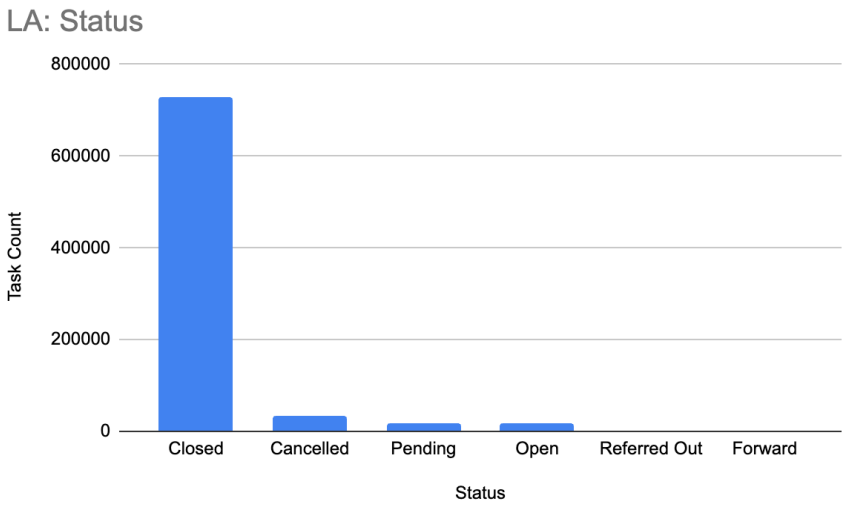
Both are dominated by closed requests (88.1% in NYC, 91.3% in LA), but their status vocabularies barely overlap: only Closed, Open, and Pending appear in both. NYC adds

workflow states (In Progress, Assigned, Started), while LA has Cancelled, Referred Out and Forward. NYC's non-closed tasks (11.9%) are mostly active work, partly because its data ends on October 5, 2026, whereas LA's (8.7%) are mostly cancellations.

Status also disagrees with the date columns. LA has about 33,650 more closed dates than Closed statuses, close to its 34,725 cancelled requests, and NYC has about 11,100 more. So we can define a completed request using both status and dates, not the closed date alone.

We can standardize status into closed, open, cancelled, and other, keeping the original value. NYC's Unspecified (53 rows) is treated as missing, and Referred Out and Forward can be categorized as other.

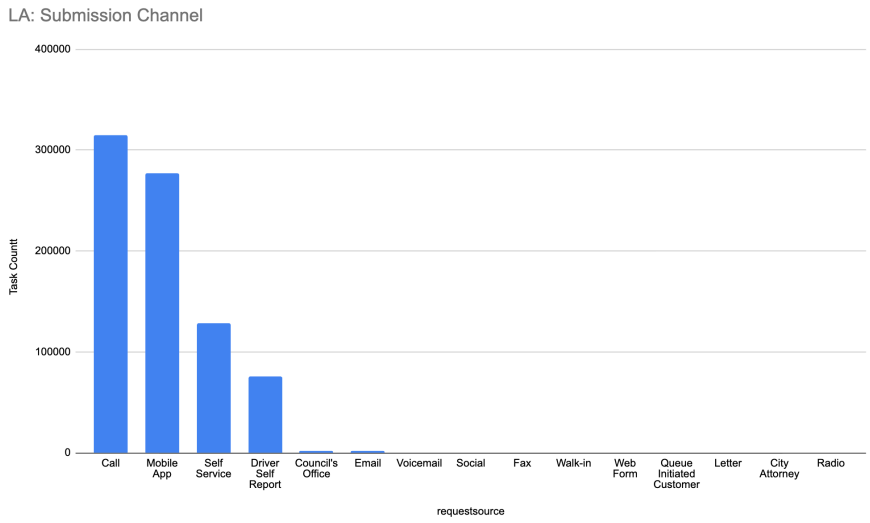


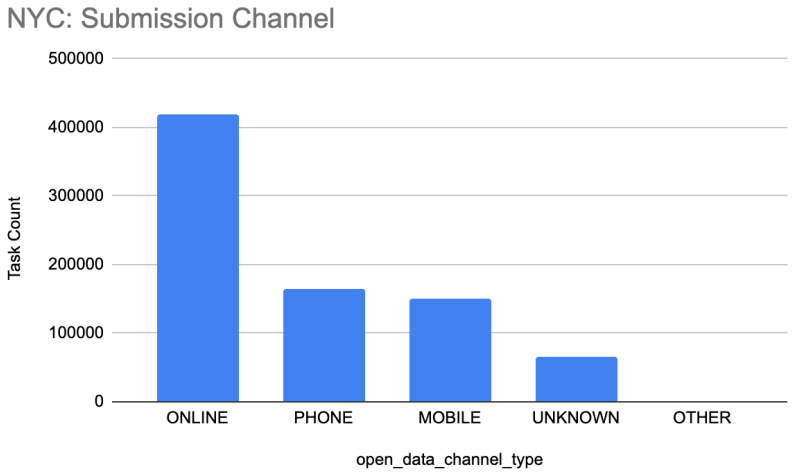


Submission Channel Distribution

NYC requests are coming in mostly through online (52.4%), with phone (20.4%) and mobile (18.8%) following next. `unknown` accounts for 8.2% and `other` takes only 0.1%. LA is led by phone calls (39.3%), then the mobile app (34.7%) and Self Service (16.1%). LA's channel column is much more granular, with 15 values, but the top four (Call, Mobile App, Self Service, Driver Self Report) cover 99.5% of requests, and the other 11 are tiny (for example, Fax has 4 records and Radio has 1).

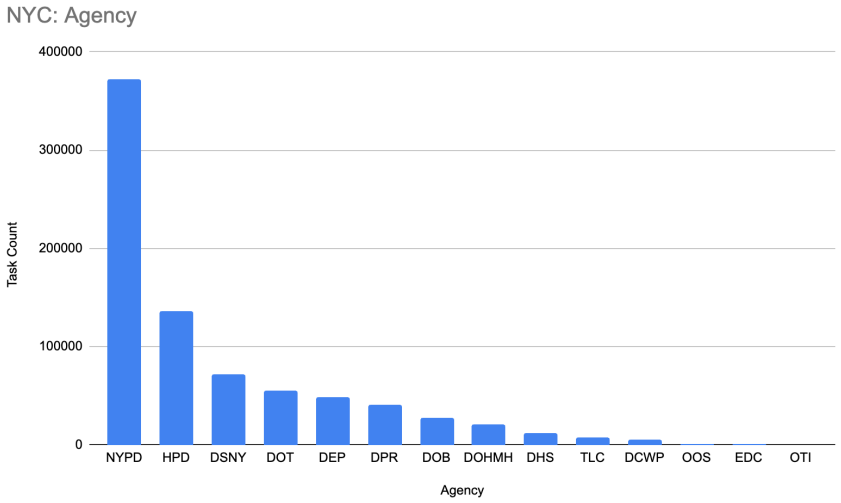
We can standardize the two categorization methods into one: Phone, Online, Mobile, and Other. NYC mapping is the simplest; `ONLINE`, `PHONE`, and `MOBILE` map directly to the proposed values, while `UNKNOWN` maps to Null. LA's `Call`, `Voicemail`, and `Queue Initiated Customer Call` map to phone. `Mobile App` maps to mobile. `Self Service`, `Email`, Web Form`, and `Social` map to online. `Driver Self Report`, `Council's Office`, `Fax`, `Letter`, `Walk-in`, `City Attorney`, and `Radio` map to Other.

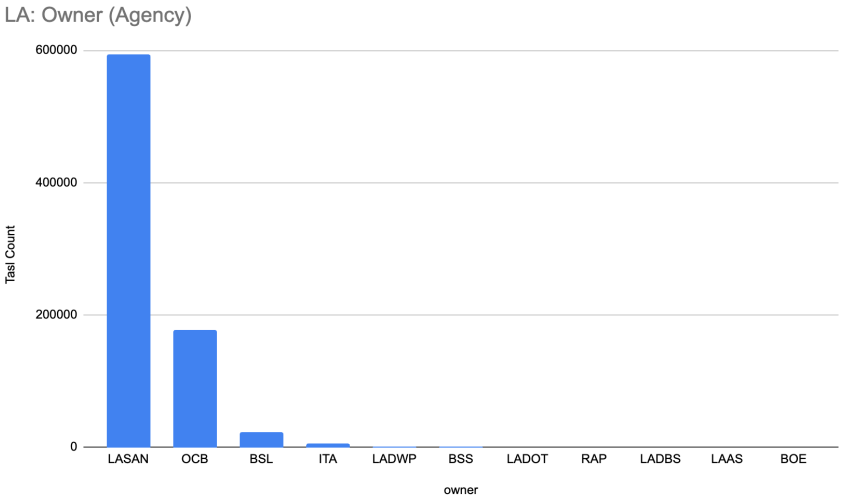




Agency/Owner Distribution

NYC requests are spread across 14 agencies but concentrated at the top agency mostly. NYPD handles 46.5% of records (372,269), followed by HPD (17.0%) and DSNY (9.0%), and the top three together account for 72.5%. LA is far more concentrated: of its 11 owners, LASAN handles 74.2% of records (593,483), and OCB handles 22.2% (177,440), so these two account for 96.4%. Every other LA owner is under 3%, and the smallest, BOE, has a single record. The pattern matches the request types. NYC's largest agencies handle enforcement and housing complaints, while LA's two largest handle pickups and cleanups. Note that OCB's count (177,440) is nearly identical to the number of Graffiti Removal requests (177,470), which suggests OCB handles graffiti, though this is an inference from the counts.



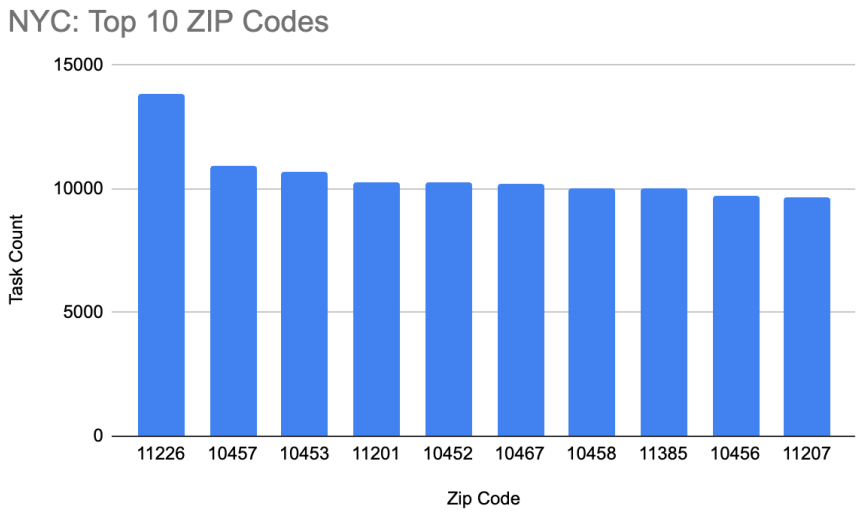


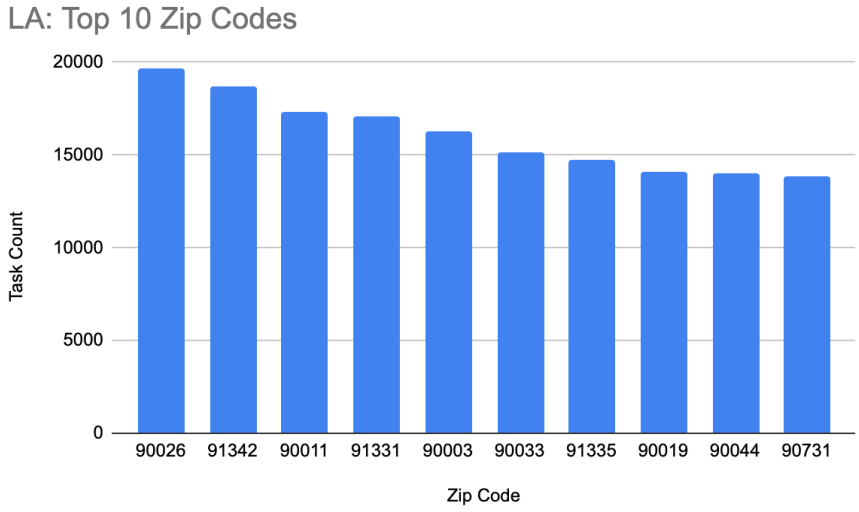
The two columns serve the same role but hold codes from different organizations, so we can't merge or compare their values across cities. We map agency and owner to a single column in the unified dataset but keep the original codes. We never group across cities, and agency-level analysis is done within each city. NYC also has agency_name with full names, but LA has no equivalent, so it stays in the cleaned NYC dataset only.

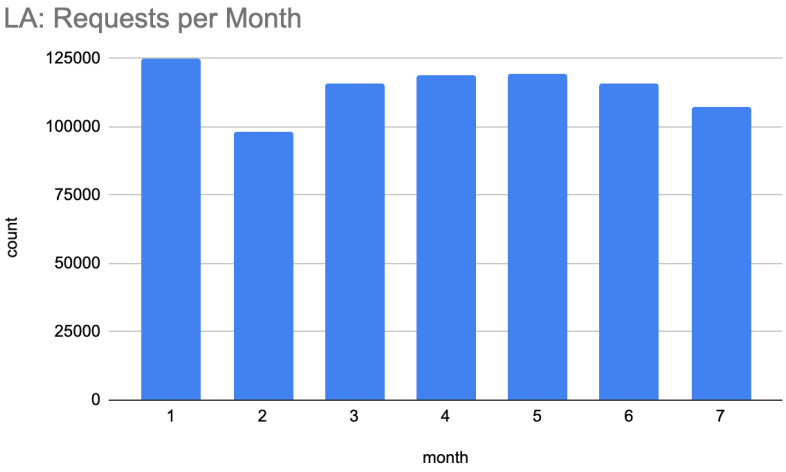
Top 10 Zip Codes

NYC's busiest ZIP codes are 11226 (13,832 requests), 10457 (10,944), and 10453 (10,670). LAs are 90026 (19,661), 91342 (18,691), and 90011 (17,299). The top 10 account for about 13.2% of NYC requests across 272 distinct ZIPs, and about 20.1% of LA requests across 142. LA's requests are therefore more concentrated in fewer ZIPs, while NYC's are spread more evenly: its #10 ZIP has 70% of the #1 count (9,635 vs. 13,832), while LA's #10 has 70% of its #1 (13,840 vs. 19,661).

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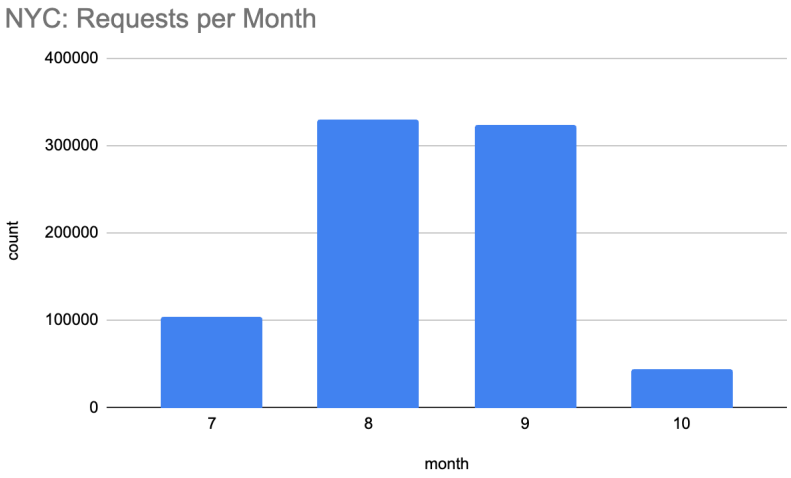




Requests Per Month

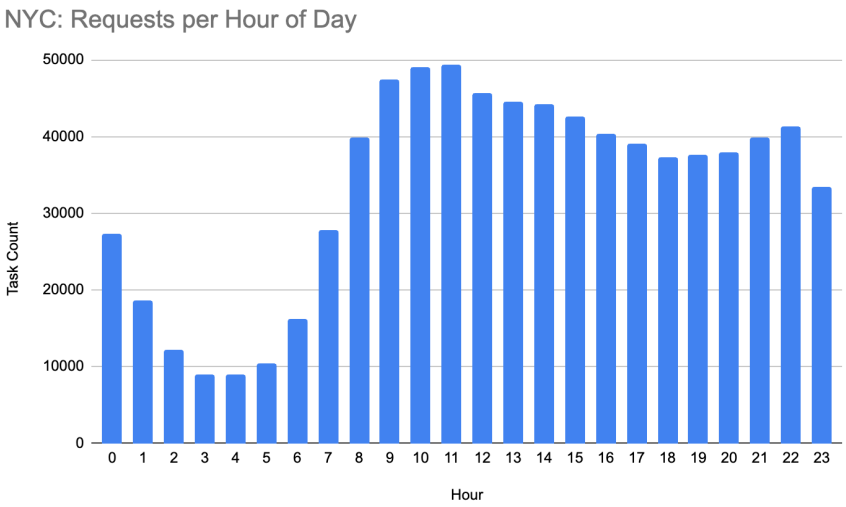
NYC requests are concentrated in two full months: August (328,905) and September (323,616) make up 81.6% of the dataset, and September is only 1.6% below August. July (103,830) and October (43,649) are low only because they are partial months. The data starts on July 22 and ends on October 5, so July covers about 10 days and October about 4. Adjusted for days covered, volume is steady at roughly 10,400 to 10,800 requests per day throughout.

LA covers January to July 2024, and its monthly totals are also fairly stable, between about 98,000 and 125,000. January is the highest (124,815) and February the lowest (98,078), but February is also the shortest month (29 days). Per day, the pattern is flatter: about 3,400 requests per day in February and 3,700 to 4,000 in other full months. July (107,119) looks lower only because the data ends on July 24, and its daily rate (about 4,460) is actually the highest.



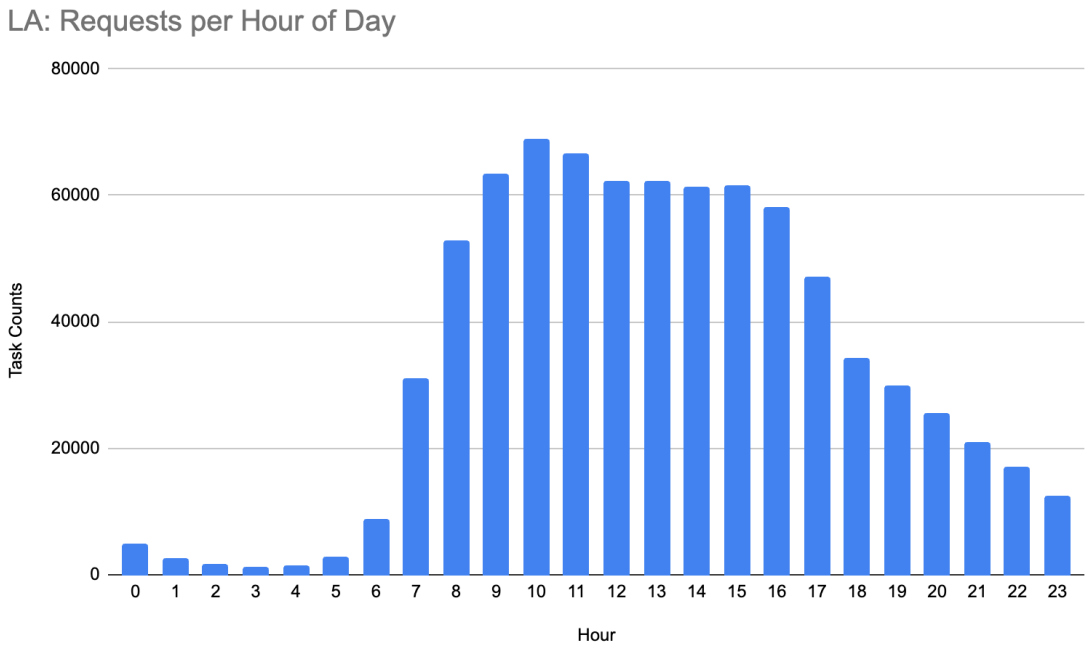
Because the partial months distort the totals, raw monthly counts shouldn't be compared across or within the cities. The cities also aren't directly comparable: NYC averages about 10,700 requests per day against LA's roughly 3,900, which is about 2.7 times higher, though both are 800,000-row extracts, so the window length is set by the exact size. The two periods don't overlap, so seasonality can't be assessed, and the cities shouldn't be compared by month.

Requests per Hour



NYC and LA both peak in the late morning, but NYC's requests are spread much more evenly across the day. NYC peaks at 11 AM (49,361) with 10 AM and 9 AM close behind, and LA peaks at 10 AM (68,932) followed by 11 AM and 9 AM. The two cities diverge outside business hours. The hours from 7 AM to 5 PM account for 73.5% of LA requests but only 53.8% of NYC's. Overnight (midnight to 5 AM) accounts for 10.8% of NYC requests and only 1.9% of LA's. NYC's quietest hour (4 AM, 8,928) still has about 18% of its peak, while LA's quietest hour (3 AM, 1,366) has only 2% of its peak.

NYC has a peak around the evening, with another peak closer to 10 PM of around 41,242. LA declines from that first peak in the morning and does not go back up, unlike the NYC data. LA mostly has requests for bulk pickup and graffiti services, and we can hypothesize that most of the services we see requested are during business hours for that reason. As we see in later parts of the EDA, NYC has mostly noise and parking complaints, which occur past business hours.



03. Exploratory Data Analysis - Key Findings

Overall Dataset Structure

The datasets are relatively clean and structured. They are made to be the same length, at 800,000 records each. Both NYC and LA datasets have a unique key (found to have no duplicates across the whole dataset: unique_key and srnumber, respectively), demonstrating that no tasks had a duplicate in either dataset.

Non-overlapping Time Periods

Both datasets cover not only a different length of time, but also cover a different year altogether. NYC goes from Jul 22 to Oct 5, 2026. LA goes from Jan 1 to Jul 24, 2024. We can use similar patterns but not a direct comparison. Apart from that, we also have partial months where the datasets do not complete the month entirely, so we must keep in mind how we analyze based on time periods.

Placeholders

Many of the columns contain placeholder values that, without analysis, might result in a value; in practice, they are actually null.

High-Null Columns

The columns with a high percentage of Null values (NYC taxi, bridge, and vehicle columns) are meant to be null. It is not necessarily a case of missing data/data inconsistencies.

Different Vocabularies

> **Status**
>
> While they operate the same function for NYC and LA, these two columns have different vocabularies. We can work to find a common mapping for both datasets to find a match across both datasets.
>
> **Source Channel**
>
> NYC has 5 values, and LA has 15. In the same way, we can find a mapping for both of the datasets to find common vocabularies. We standardize phone, online, mobile, and other.

Request Types

184 NYC types vs. 12 LA types. LA's top 2 types make up 69% of records, versus 26% for NYC. Looking specifically at LA, we can match any of the columns with the types in NYC (seeing as LA has the smallest number of categories) and create a category `other` for those we cannot match.

04. Data Quality & Cleaning Decisions

The EDA showed us that both datasets were relatively structured and of good quality, with a few columns needing attention before merging the datasets. The main issues were conflicts between dates, statuses, and resolution times, invalid ZIP codes, and placeholder values. Our approach to data cleaning was to correct or replace values while retaining as much information and as many records as we could. All requests were retained, and the original sources were also left unchanged.

4.1 Closed Dates and Status Conflicts

The most important inconsistencies in the two datasets occurred between the creation date, closed date, and status fields. We checked these together and identified five related conditions.

Condition before cleaning	NYC	LA	Decision
1. Closed status without a closed date	4	1,991	Keep the status and leave the date null.
2. Closed date before the creation date	626	0	Replace the invalid closed date with null.
3. Closed date present, but status not Closed	11,109	35,643	Check the status before deciding whether to change it.
4. Eligible for conversion to Closed	11,109	0	Update cleaned_status to Closed.
5. Cancellation, referral, or forwarding status with a closed date	0	35,643	Preserve the existing outcome.

These conditions can overlap, so the counts should not be added together.

4.1.1 Closed status without a closed date

We left the missing closed dates for 4 NYC and 1,991 LA requests as null. Replacing a missing date with an average would introduce an estimated closure time and could affect

later analysis. In this case, when analyzing requests that require closed dates, these null values can be excluded. This ensures that records without usable closed dates are not included in that analysis while preserving the requests' other information.

4.1.2 Closed date before the creation date

We replaced the 626 NYC closed dates that occurred before creation with null. These dates cannot describe a valid sequence of events, but we did not have enough information to determine the correct dates. We kept the requests because the remaining information could still be useful.

4.1.3 Closed date present, but status not Closed

This conflict mainly involved the various statuses of requests that had a closed date. The decision was based on two types of statuses: those reflecting administrative outcomes and those more likely to reflect an error.

The administrative statuses included Cancelled, Referred Out, and Forward, which indicate that the request may have been cancelled or may no longer be handled by the corresponding local agency. These describe different outcomes and should not automatically become Closed. Thus, no LA statuses were changed by this rule but were all preserved.

The potentially erroneous statuses included active and unspecified statuses, which we believed could be inconsistent with the recorded closed dates. For active or unspecified statuses, we used the recorded closure date as the stronger indication that the request had closed. Thus, the status of those 11,109 requests from NYC are converted to Closed.

4.2 Invalid ZIP Codes

We checked whether the ZIP codes were exactly five digits long and found 1 invalid NYC ZIP value (No) and 39 invalid LA ZIP values: 38 instances of 0 and 1 instance of VE, 0. We replaced these values with null. All requests were retained, and existing missing ZIP values were left missing. In addition, ZIP codes were stored as strings because they

identify an area rather than represent a quantity.

Zip codes	Dataset	count
“No”	NYC	1
“0”	LA	38
“VE, 0”	LA	1

4.3 Placeholder Values

NYC originally had 53 Unspecified statuses. The date rule changed 46 of these to Closed, leaving 7 without a known status. We replaced those 7 values with null in cleaned_status. We also replaced the 65,687 UNKNOWN submission_channels with null. However, we retained OTHER submission_channels because it represents a reported category outside the main channels, rather than an unknown channel.

The general cleaning rules apply the same principle to documented placeholders in NYC city, borough, police precinct, descriptor, location type, and community board fields. In LA, a house number of zero is treated as missing because it does not identify a building. These replacements make missing information easier to identify without inventing values.

4.4 Missing Values

After running the EDA, we found that many columns had a high percentage of missing values. We did not remove columns simply because they had a high percentage of missing values. As discussed in the EDA, some fields apply only to certain requests. NYC’s taxi and bridge fields, for example, are not expected to contain values for every complaint. Similarly, LA’s mobile operating system field is relevant only to requests submitted through a mobile device.

Therefore, we retained all source columns and left missing values unfilled. This is because, after mapping the schemas between the two datasets, we realized that most of

those columns with missing values could not be included in the final dataset anyway. The only two that could be mapped were the street name field, which is redundant because the information is also available in the address field, and the closed date field, whose missing values may be due to ongoing cases or various other reasons. Thus, all columns were preserved for city-specific analysis.

4.5 Text, Date, and Location Consistency

The general cleaning rules address smaller formatting differences so that equivalent values do not appear as separate categories.

Area	Cleaning rule	Reason
Text fields	Trim leading and trailing spaces, while preserving source status.	Prevent spaces from creating artificial differences.
Date fields	Convert the listed date columns to timestamps.	Support consistent ordering and elapsed-time calculations.
NYC city labels	Use uppercase and map MANHATTEN to MANHATTAN, and NYand NEW YORK CITY to NEW YORK.	Combine observed spelling and naming variants.
NYC complaint types	Use uppercase.	Make capitalization consistent before category mapping.
LA street suffixes	Map AV to AVE, RD to ROAD, LN to LANE, and WY to WAY.	Combine the documented abbreviation variants.
Coordinates	Replace zero or globally invalid latitude and longitude values with null.	Remove unusable coordinate values.
LA map pages	Replace the documented values 4641, 4723, and 4724 with null.	Exclude the specific map-page outliers identified for cleaning.

The NYC city rules also clear a documented list of unrelated place names and junk values while retaining valid neighborhood names. We did not use a broad rule that removes every value other than the five borough names, since that could remove useful local names.

For coordinates, the notebook uses global latitude and longitude limits rather than a city boundary. This removes impossible values. We also leave LA's 'addressverified = M' unchanged because its meaning was not established. An unfamiliar code alone is not enough reason to replace it.

4.6 Duplicates

Both datasets have unique request IDs. However, the EDA found 261 surplus NYC records across 217 groups that matched on every other source column. No equivalent duplicates were found in LA.

We retained these records because different IDs may represent separate submissions, even when the recorded details are identical. The dataset does not establish whether they were created accidentally or submitted more than once.

The notebook adds is_suspected_duplicate to identify the surplus NYC records. Within each matching group, the record with the lowest ID is left unflagged and the remaining records are flagged. This is a consistent reference rule; it does not prove which request was the original. The comparison uses the source values so that cleaning does not create artificial matches.

4.7 Resolution Hours

We further calculate the elapsed resolution_hours as the difference between the creation dates and closed dates. The value is only calculated when both timestamps exist and closure date is not before creation. Otherwise, the duration remains null. We also added the 'is_valid_resolution' flag which identifies records with a timestamp pair. This flag checks the dates, not whether the service was successfully completed. Thus a cancelled or referred request can still have a valid elapsed time.

4.8 Validation Check

At the end, we ran the validation checks, and the outputs confirm that the main date, status, ZIP, and placeholder cleaning steps retained 800,000 requests and 800,000 distinct IDs in each city. The checks also confirm that the original statuses were preserved and

that no invalid non-missing ZIP formats remained.

Verified result	NYC	LA
Requests retained	800,000	800,000
Distinct request IDs	800,000	800,000
Missing ZIP values after ZIP cleaning	6,882	926
Closed requests without usable closed dates	630	1,991

05. Schema Comparison & Standardization

Schema Analysis

NYC 311 Schema (44 columns)

timestamp (4): created_date, closed_date, due_date, resolution_action_updated_date

integer (4): unique_key, council_district, x_coordinate_state_plane, y_coordinate_state_plane

long (1): bbl

double (2): latitude, longitude

string (33): agency, agency_name, complaint_type, descriptor, descriptor_2, location_type, incident_zip, incident_address, street_name, cross_street_1, cross_street_2, intersection_street_1, intersection_street_2, address_type, city, landmark, facility_type, status, resolution_description, community_board, police_precinct, borough, open_data_channel_type, park_facility_name, park_borough, vehicle_type, taxi_company_borough, taxi_pick_up_location, bridge_highway_name, bridge_highway_direction, road_ramp, bridge_highway_segment, location

LA MyLA311 Schema (34 columns)

timestamp (4): createddate, updateddate, servicedate, closeddate

integer (5): housenumber, tbmpage, tbmrow, cd, nc

double (2): latitude, longitude

string (23): srnumber, actiontaken, owner, requesttype, status, requestsource, createdbyuserorganization, mobileos, anonymous, assignto, addressverified, approximateaddress, address, direction, streetname, suffix, zipcode, location, tbmcolumn, apc, cdmember, ncname, policeprecinct

06. Building the Unified Dataset

Concept	NYC_column	NYC_type	LA_column	LA_type
id	unique_key	int	srnumber	string
created	created_date	timestamp	createddate	timestamp
closed	closed_date	timestamp	closeddate	timestamp
updateddate	resolution_action_updated_date	timestamp	updateddate	timestamp
status	status	string	status	string
request type	complaint_type	string	requesttype	string
channel	open_data_channel_type	string	requestsource	string
zip	incident_zip	string	zipcode	string
agency	agency	string	owner	string
adress	incident_address	string	address	string
streetname	street_name	string	streetname	string
policeprecint	police_precinct	string	policeprecinct	string
requestedby	open_data_channel_type	string	requestsource	string
latitude	latitude	double	latitude	double
longitude	longitude	double	longitude	double
location	location	string	location	string

In order to begin merging the two datasets, we first had to match corresponding columns from each dataset that had the same meaning. That is the matching that we see above. Both columns from their corresponding datasets have the same concept, as we have seen in the EDA. However, many have different vocabularies/categories, so we must standardize them in order to merge the two datasets.

Column-by-Column Merging

> **`uid`:** We create another unique id to avoid having any of the unique keys from the two datasets matching with each other (which will most likely not happen, as they follow different formats, but we create out of abundance of caution). The format followed is `uid` = source + unique key.
>
> **`source`:** In order to differentiate between the two datasets, we included a column called `source`, that is filled with either NYC or LA, corresponding to their respective cities.
>
> **`id`:** Copied directly from the unique keys that each dataset contained, and cast to string format.
>
> **`created`:** Copied directly from the corresponding columns in both of the datasets.

`closed`: Copied directly from the corresponding columns in both of the datasets.

`updated_date`: Copied directly from the corresponding columns in both of the datasets.

`status`: Had to use mapping in order to standardize this column across the two different datasets. The mapping used was of the same principle as that of the EDA. We had 4 categories total: closed, open, cancelled, and other. Mapping is as follows: {"closed": "closed", "open": "open", "pending": "open", "in progress": "open", "assigned": "open", "started": "open", "cancelled": "cancelled", "referred out": "other", "forward": "other"}. We used this mapping for both NYC and LA, then created the standardized status column.

`status_detail`: In order to preserve the detail in the NYC and LA datasets, we kept the cleaned_status column from their respective datasets here.

`request_type`: We used the same principle as for status, but with individual mappings, as the range of values is much larger for each dataset.

> NYC mapping: {"graffiti": "Graffiti Removal", "encampment": "Homeless Encampment", "dead animal": "Dead Animal Removal", "street light condition": "Streetlight Issue", "illegal dumping": "Illegal Dumping Pickup", "water conservation": "Report Water Waste"}
>
> LA mapping: {"bulky items": "Bulky Items", "dead animal removal": "Dead Animal Removal", "electronic waste": "Electronic Waste", "feedback": "Feedback", "graffiti removal": "Graffiti Removal", "homeless encampment": "Homeless Encampment", "illegal dumping pickup": "Illegal Dumping Pickup", "metal/household appliances": "Metal/Household Appliances", "multiple streetlight issue": "Streetlight Issue", "single streetlight issue": "Streetlight Issue", "report water waste": "Report Water Waste", "other": "Other"}

After the mapping was conducted to standardize the column, we can finalize the request_type column to compare the two datasets.

`request_type_original`: In order to preserve the original data, as done with status, we keep a column for the original categories.

`channel`: Channel, as seen in the EDA, had a lot more values for LA, compared

to NYC. The mapping followed the principle shown in the EDA section as well.

> NYC mapping: {"phone": "Phone", "online": "Online", "mobile": "Mobile", "other": "Other"}
>
> LA mapping: {"call": "Phone", "voicemail": "Phone", "queue initiated customer call": "Phone", "mobile app": "Mobile", "self service": "Online", "email": "Online", "web form": "Online", "social": "Online", "driver self report": "Other", "council's office": "Other", "fax": "Other", "letter": "Other", "walk-in": "Other", "city attorney": "Other", "radio": "Other"}

`channel_original`: Following the same principle as before, we keep a column with all original values.

`zip`: Zip codes had matching formats across the two datasets, we simply merged the values together into this column.

`agency`: Because the agencies/owners are different between the two datasets (seeing as they are two different cities), we simply standardized the capitalization across the two datasets and merged them.

`address`: The two datasets followed a similar format, so we can simply merge the two datasets together in this column.

`street_name`: The two datasets followed a similar format, so we can simply merge the two datasets together in this column.

`police_precinct`: Following the same basic idea of agency, we found no way to standardize across datasets, since they are following different formats. Additionally, no record from either dataset will hold the police precinct of the opposite dataset; hence, there is no dire need to standardize the vocabularies.

`latitude`: The two datasets followed the same format; we can simply merge the two datasets together in this column.

`longitude`: The two datasets followed the same format; we can simply merge the two datasets together in this column.

`location`: The two datasets followed the same format; we can simply merge the two datasets together in this column.

`resolution_hours`: This derived column denotes the total hours taken for an

issue/task to be resolved. The format is the same across both datasets, as it is calculated directly from the closed and open dates.

`is_valid_resolution`: As depicted in the EDA, some records had closed dates before their open dates. In the data cleaning, we input the closing date as null, and create a flag column to provide a way to avoid them when performing analysis on the resolution hours columns.

`is_suspected_duplicate`: We noted in the EDA that there exist columns that are completely matching in all attributes but their unique id's. We cannot conclude they are duplicates, but we flag them for future reference.

07. Conclusion

In Phase 1, we were able to understand the structure and limitations of the NYC and LA 311 datasets. The EDA showed that both datasets were relatively clean and structured, with 800,000 records each and unique request IDs. However, they covered different time periods and had different vocabularies for statuses, submission channels, and request types. We also found placeholder values and columns with high percentages of missing values, many of which reflected information that applied only to specific requests. These findings helped us identify what needed cleaning and what limitations we should consider when comparing the two cities.

Furthermore, we established cleaning rules that focused on preserving information and records rather than removing entire columns or requests unnecessarily. The merging and integration of the two datasets were completed successfully, and the final mapped schema can also be found in the DBMS Google spreadsheet [1]. Overall, we now have a clearer understanding of how to work with the integrated data and can build further analyses on the foundation provided by this merged dataset.